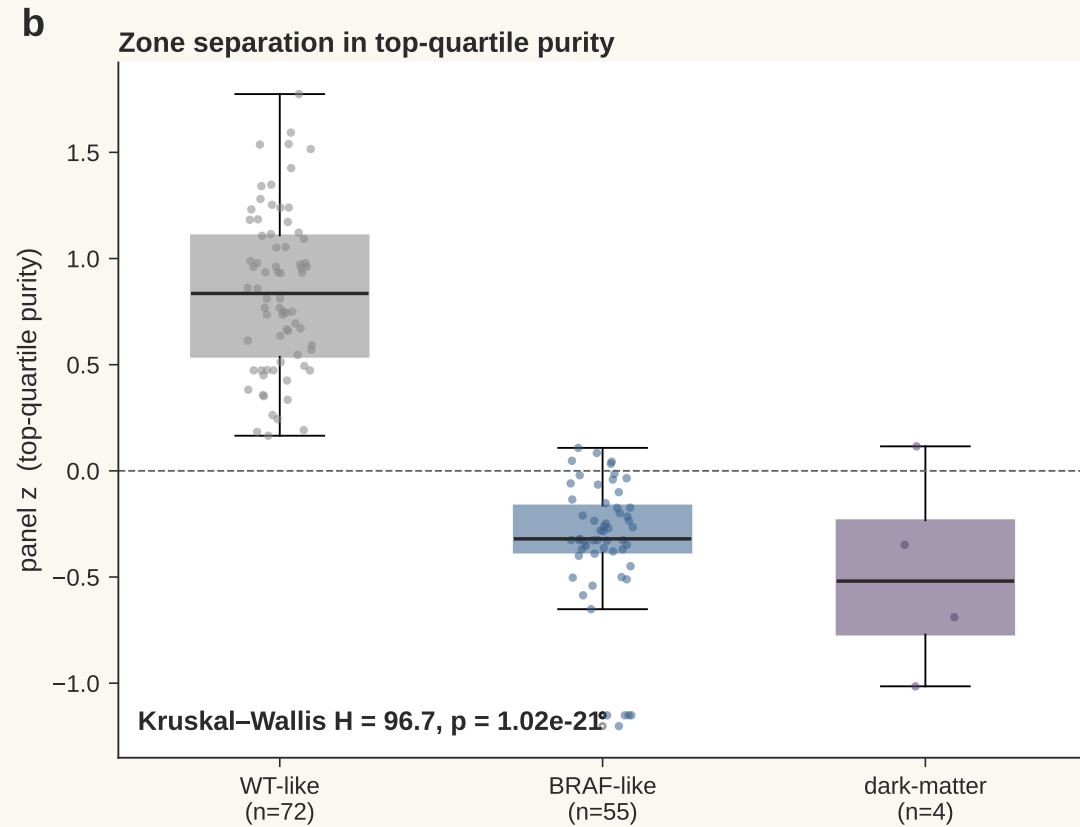
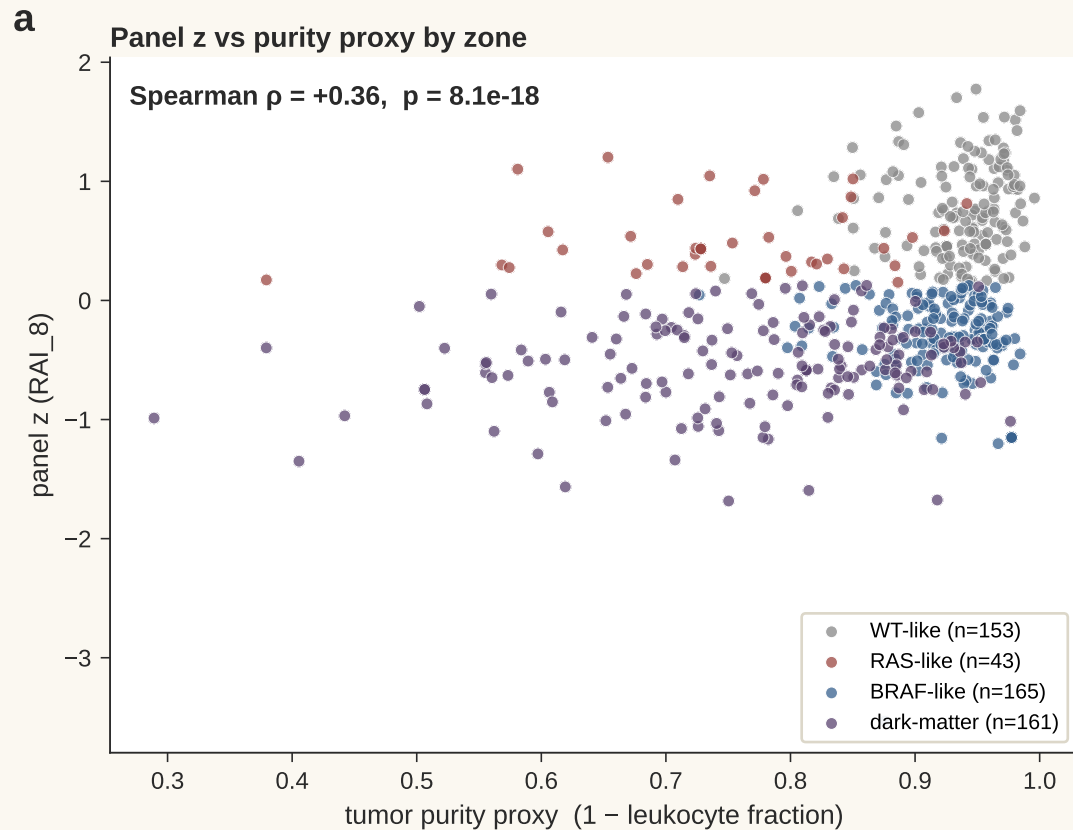


Panel z is not driven by tumor purity (TCGA-THCA, n = 522)



Panel z–purity Spearman $\rho = +0.36$; zone separation persists in high-purity tumours (KW $p = 1.02e-21$).